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Practical 8

Saving the JWT Token:

```
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\student-records\student-records> $token = "eyJhbGciOiJIUzI1NiJ9.eyJzdWIiOiJwcm15YW5rYSIsIm1hdCI6MTc5MDc..."
PS C:\Users\DELL\OneDrive\Documents\S DBE&DBD\Practical\student-records\student-records> $headers = @{
>>     Authorization = "Bearer $token"
>> }
```

Folder Structure:

Week-8_Practical_Student_Records/

```
|
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|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   | └─ StudentRecordsApplication.java
|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   | └─ controller/
|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   | └─ dto/
|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   | └─ entity/
|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   | └─ exception/
|   |   |   |   |   |   |   |
|   |   |   |   |   |   |   | └─ repository/
```

```

| | | └─ security/
| | | └─ service/
| | |
| | └─ resources/
| | └─ application.properties
| |
| └─ target/

```

Codes:

1. Main Application

□ StudentRecordsApplication.java

```

package com.example.student_records;
import org.springframework.boot.SpringApplication; import
org.springframework.boot.autoconfigure.SpringBootApplication;
@SpringBootApplication public class
StudentRecordsApplication {

    public static void main(String[] args) {
        SpringApplication.run(StudentRecordsApplication.class, args);
    }

}

```

2. Entity

□ Student.java

```

package com.example.student_records.entity;

import org.springframework.data.annotation.Id;
import org.springframework.data.mongodb.core.mapping.Document;
@Document(collection="students")
public class Student {

    @Id
    private String id;
    private String name;
    private Integer age;
}

```

```

private String course;
private String email;
    public String getId()
{
    return id;
}
    public void
setId(String id) {
this.id = id;
}
    public String
getName() {
return
name;
}
    public void setName(String
name) {
this.name = name;
}
    public Integer
getAge() {
return
age;
}
    public void
setAge(Integer age) {
this.age = age;
}
    public String getCourse() {
return course;
}
    public void setCourse(String
course) {
this.course = course;
}
    public String
getEmail() {
return
email;
}
    public void setEmail(String
email) {
this.email = email;
}
}

```

□ User.java

```

package com.example.student_records.entity;
import org.springframework.data.annotation.Id; import
org.springframework.data.mongodb.core.index.Indexed; import
org.springframework.data.mongodb.core.mapping.Document;

@Document(collection = "users")
public class User {

    @Id
    private String id;

    @Indexed(unique = true)
    private String username;
    private String
password;
    public String getId()
{
    return id;
}

```

```

    }      public void
setId(String id) {
this.id = id;
    }      public String
getUsername() {      return
username;
    }
    public void setUsername(String username) {
this.username = username;
    }      public String
getPassword() {      return
password;
    }
    public void setPassword(String password) {
this.password = password;
    }
}

```

3. DTO

□ StudentRequest.java

```

package com.example.student_records.dto;
import jakarta.validation.constraints.Email;
import jakarta.validation.constraints.Min;
import
jakarta.validation.constraints.NotBlank;
import jakarta.validation.constraints.NotNull;
public class StudentRequest
{

    @NotBlank(message = "name is required")
private String name;

    @NotNull(message = "age is required")
    @Min(value = 16, message = "age must be at least 16")
private Integer age;

    @NotBlank(message = "course is required")
private String course;

    @NotBlank(message = "email is required")
    @Email(message = "a valid email is required")
private String email;
    public String getName()
{
    return name;
}

```

```

    }    public void setName(String
name) {    this.name = name;
    }
    public Integer getAge() {
return age;
    }    public void
setAge(Integer age) {
this.age = age;
    }    public String
getCourse() {    return
course;
    }    public void setCourse(String
course) {    this.course = course;
    }    public String
getEmail() {    return
email;
    }    public void setEmail(String
email) {    this.email = email;
    }
}

```

□ RegisterRequest.java

```

package com.example.student_records.dto;
import
jakarta.validation.constraints.NotBlank;
import jakarta.validation.constraints.Size;
public class RegisterRequest
{

    @NotBlank    private
String username;

    @NotBlank
    @Size(min = 6, message = "password must be at least 6 characters")
private String password;
    public String getUsername()
{
    return username;
}
    public void setUsername(String
username) {    this.username = username;
    }    public String
getPassword() {    return
password;
    }    public void setPassword(String
password) {    this.password = password;
    }
}

```

□ LoginRequest.java package

```
com.example.student_records.dto; import
jakarta.validation.constraints.NotBlank
;

public class LoginRequest
{
    @NotBlank    private
String username;

    @NotBlank    private
String password;
    public String getUsername()
{
    return username;
}
    public void setUsername(String
username) {
    this.username = username;
}
    public String
getPassword() {
    return
password;
}
    public void setPassword(String
password) {
    this.password = password;
}
}
```

4. Repository

- StudentRepository.java

```
package com.example.student_records.repository;

import com.example.student_records.entity.Student; import
org.springframework.data.mongodb.repository.MongoRepository;
    public interface StudentRepository extends MongoRepository<Student,
String>
{
}
}
```

- UserRepository.java

```
package com.example.student_records.repository;
import
com.example.student_records.entity.User;
```

```

import org.springframework.data.mongodb.repository.MongoRepository;
import
java.util.Optional;
public interface
UserRepository extends
MongoRepository<User,
String> {
    Optional<User> findByUsername(String username);
}

```

5. Service

□ StudentService.java

```

package com.example.student_records.service;
import com.example.student_records.dto.StudentRequest; import
com.example.student_records.entity.Student; import
com.example.student_records.repository.StudentRepository;
import org.springframework.stereotype.Service;
import java.util.List;
import
java.util.Optional;
@Service public class
StudentService {
    private final StudentRepository
repository;
    public StudentService(StudentRepository repository)
{
        this.repository = repository;
    }
    public Student
create(StudentRequest r) {

        Student s = new Student();

        apply(s, r);

        return repository.save(s);
    }
    public List<Student>
getAll() {
        return
repository.findAll();
    }

    public Optional<Student> getById(String id) {
return repository.findById(id);
    }
    public Optional<Student> update(String id, StudentRequest
r) {
        return repository.findById(id).map(s -> {

```

```

        apply(s,
r);
        return
repository.save(s);
    });
}    public boolean
delete(String id) {
    if (!repository.existsById(id))
{
        return false;
    }
    repository.deleteById(id);
    return
true;
}    private void apply(Student s,
StudentRequest r) {

    s.setName(r.getName());
    s.setAge(r.getAge());
    s.setCourse(r.getCourse());
    s.setEmail(r.getEmail());
}
}

```

6. Controller

□ StudentController.java

```

package com.example.student_records.controller;

import com.example.student_records.dto.StudentRequest;
import com.example.student_records.entity.Student; import
com.example.student_records.service.StudentService;

import jakarta.validation.Valid;
import org.springframework.http.ResponseEntity;
import
org.springframework.web.bind.annotation.*;
import
java.util.List;

@RestController
@RequestMapping("/students")
public class StudentController {
    private final StudentService
service;
    public StudentController(StudentService service)
{
        this.service = service;

```

```

    }

    @PostMapping      public
    ResponseEntity<Student> create(
        @Valid @RequestBody StudentRequest r) {
        return
    ResponseEntity
        .status(201)
        .body(service.create(r));
    }

    @GetMapping      public
    List<Student> getAll() {
    return service.getAll();
    }

    @GetMapping("/{id}")      public
    ResponseEntity<Student> getById(
        @PathVariable String id) {
        return
    service.getById(id)
        .map(ResponseEntity::ok)
        .orElse(ResponseEntity.notFound().build());
    }

    @PutMapping("/{id}")
    public ResponseEntity<Student> update(
        @PathVariable String id,
        @Valid @RequestBody StudentRequest r) {
        return service.update(id,
    r)
        .map(ResponseEntity::ok)
        .orElse(ResponseEntity.notFound().build());
    }

    @DeleteMapping("/{id}")      public
    ResponseEntity<Void> delete(
        @PathVariable String id) {

        return service.delete(id)
            ? ResponseEntity.noContent().build()
            : ResponseEntity.notFound().build();
    }
}

```

□ AuthController.java

```

package com.example.student_records.controller;
import com.example.student_records.dto.LoginRequest; import
com.example.student_records.dto.RegisterRequest; import
com.example.student_records.entity.User; import
com.example.student_records.repository.UserRepository;
import com.example.student_records.security.JwtUtil;
import
jakarta.validation.Valid;
import
org.springframework.http.ResponseEntity;
import org.springframework.security.authentication.AuthenticationManager;
import
org.springframework.security.authentication.UsernamePasswordAuthentication
Token; import
org.springframework.security.crypto.password.PasswordEncoder;
import
org.springframework.web.bind.annotation.*;
import
java.util.Map;

@RestController
@RequestMapping("/auth")
public class AuthController {
    private final UserRepository users;
    private final PasswordEncoder encoder;
    private final AuthenticationManager manager;
    private final JwtUtil jwt;

    public AuthController(
UserRepository users,
        PasswordEncoder encoder,
        AuthenticationManager manager,
        JwtUtil jwt) {

        this.users = users;
this.encoder = encoder;
this.manager = manager;
this.jwt = jwt;
    }

    @PostMapping("/register")
    public ResponseEntity<?> register(
        @Valid @RequestBody RegisterRequest r) {        if

        (users.findByUsername(r.getUsername()).isPresent()) {

            return ResponseEntity
.badRequest()
                .body(Map.of(
                    "error",

```

```

        "username already taken"
    ));
}

User user = new User();

user.setUsername(r.getUsername());
user.setPassword(
    encoder.encode(r.getPassword())
);
users.save(user);
return
ResponseEntity
    .status(201)
    .body(Map.of(
        "message",
        "registered successfully"
    ));
}

@PostMapping("/login")
public ResponseEntity<?> login(
    @Valid @RequestBody LoginRequest r) {

    manager.authenticate(
        new
        UsernamePasswordAuthenticationToken(
            r.getUsername(),
            r.getPassword()
        )
    );

    return ResponseEntity.ok(
        Map.of(
            "token",
            jwt.generateToken(r.getUsername())
        )
    );
}
}

```

7. Security

□ SecurityConfig.java

```

package com.example.student_records.security;
import org.springframework.context.annotation.Bean; import
org.springframework.context.annotation.Configuration;

```



```

        .authorizeHttpRequests(auth ->
auth
            .requestMatchers("/auth/**")
            .permitAll()

            .anyRequest()
            .authenticated()
        )

        .addFilterBefore(
jwtAuthFilter,
            UsernamePasswordAuthenticationFilter.class
        )

        .build();
    }
}

```

□ JwtUtil.java

```

package com.example.student_records.security;

import io.jsonwebtoken.*; import
io.jsonwebtoken.security.Keys;
import
org.springframework.beans.factory.annotation.Value;
import org.springframework.stereotype.Component;
import javax.crypto.SecretKey; import
java.nio.charset.StandardCharsets;
import java.util.Date;
@Component public
class JwtUtil {

    @Value("${jwt.secret}")
private String secret;

    @Value("${jwt.expiration-ms}")
private long expirationMs;
    private SecretKey key()
{
        return Keys.hmacShaKeyFor(
secret.getBytes(StandardCharsets.UTF_8)
        );
    }
    public String generateToken(String
username) {

        Date now = new Date();

```

```

        return Jwts.builder()
            .setSubject(username)
                .setIssuedAt(now)
            .setExpiration(
                Date(now.getTime() + expirationMs)
            )
                .signWith(key(), SignatureAlgorithm.HS256)
                .compact();
    }
    public String extractUsername(String
token) {

        return Jwts.parserBuilder()
            .setSigningKey(key())
            .build()
            .parseClaimsJws(token)
            .getBody()
            .getSubject();
    }
    public boolean isValid(String
token) { try {

        Jwts.parserBuilder()
            .setSigningKey(key())
            .build()
            .parseClaimsJws(token);

        return true;

    } catch (JwtException | IllegalArgumentException e) {
        return
false;
    }
}
}

```

□ JwtAuthFilter.java

```

package com.example.student_records.security;
import jakarta.servlet.*;
import
jakarta.servlet.http.*;
import
org.springframework.security.authentication.UsernamePasswordAuthentication
T
oken;
import
org.springframework.security.core.context.SecurityContextHolder;
import
org.springframework.security.core.userdetails.UserDetailsService;
import
org.springframework.security.web.authentication.WebAuthenticationDetailsSo
u rce;

```

```

import org.springframework.stereotype.Component;
import org.springframework.web.filter.OncePerRequestFilter;
import
java.io.IOException;

@Component public class JwtAuthFilter extends
OncePerRequestFilter {

    private final JwtUtil jwtUtil;
    private final UserDetailsService userDetailsService;

    public JwtAuthFilter(
        JwtUtil jwtUtil,
        UserDetailsService userDetailsService) {
        this.jwtUtil =
jwtUtil;
        this.userDetailsService =
userDetailsService;    }

    @Override
    protected void doFilterInternal(
HttpServletRequest request,
        HttpServletResponse response,
FilterChain chain) throws
ServletException, IOException {
        String header = request.getHeader("Authorization");
        if (header != null && header.startsWith("Bearer "))
    {
        String token = header.substring(7);
        if (jwtUtil.isValid(token))
    {

                System.out.println("JWT VALID");
                String username =
jwtUtil.extractUsername(token);
                var details =
userDetailsService.loadUserByUsername(username);
                var authentication =
new UsernamePasswordAuthenticationToken(
details,                                null,
details.getAuthorities()
                );

        authentication.setDetails(
                new WebAuthenticationDetailsSource()
                    .buildDetails(request)
                );

        SecurityContextHolder
            .getContext()

```

```

        .setAuthentication(authentication);
    } else {

        System.out.println("JWT INVALID");
    }
}

chain.doFilter(request, response);
}
}

```

□ CustomUserDetailsService.java

```

package com.example.student_records.security;
import
com.example.student_records.repository.UserRepository;
import
org.springframework.security.core.userdetails.*;
import org.springframework.stereotype.Service;
@Service public class CustomUserDetailsService implements
UserDetailsService {
    private final UserRepository
userRepository;
    public CustomUserDetailsService(UserRepository userRepository)
{
        this.userRepository = userRepository;
    }

    @Override    public UserDetails
loadUserByUsername(String username)                throws
UsernameNotFoundException {
        var user =
userRepository
                .findByUsername(username)
                .orElseThrow(() ->                new
UsernameNotFoundException("User not found"));

        return User.withUsername(user.getUsername())
                .password(user.getPassword())
                .authorities("USER")
                .build();
    }
}

```

8. Exception Handling

□ GlobalExceptionHandler.java

```

package com.example.student_records.exception;

import org.springframework.http.ResponseEntity; import
org.springframework.web.bind.MethodArgumentNotValidException;
import org.springframework.web.bind.annotation.ExceptionHandler;
import
org.springframework.web.bind.annotation.RestControllerAdvice;
import
java.util.HashMap;
import java.util.Map;
@RestControllerAdvice public class
GlobalExceptionHandler {

    @ExceptionHandler(MethodArgumentNotValidException.class)
    public ResponseEntity<Map<String, Object>> handleValidation(
        MethodArgumentNotValidException ex) {

        Map<String, String> errors = new HashMap<>();
        ex.getBindingResult()
            .getFieldErrors()                .forEach(error ->
            errors.put(
                error.getField(),
                error.getDefaultMessage()
            )
        );

        Map<String, Object> body = new HashMap<>();
        body.put("error",
            "VALIDATION_ERROR");
        body.put("details", errors);
        return
            ResponseEntity.badRequest().body(body);
    }
}

```

9. Configuration

```

application.properties

spring.application.name=student-records

spring.data.mongodb.uri=mongodb://localhost:27017/student_records

jwt.secret=replace-this-with-a-long-random-secret-at-least-32-characters

jwt.expiration-ms=3600000

server.port=8080

```

10. Maven Configuration

□ pom.xml

```
<?xml version="1.0" encoding="UTF-8"?>
<project xmlns="http://maven.apache.org/POM/4.0.0"
  xmlns:xsi="http://www.w3.org/2001/XMLSchema-instance"
    xsi:schemaLocation="http://maven.apache.org/POM/4.0.0
  https://maven.apache.org/xsd/maven-4.0.0.xsd">

  <modelVersion>4.0.0</modelVersion>

  <parent>
    <groupId>org.springframework.boot</groupId>
    <artifactId>spring-boot-starter-parent</artifactId>
    <version>3.5.6</version>
    <relativePath/>
  </parent>

  <groupId>com.example</groupId>
  <artifactId>student-records</artifactId>
  <version>0.0.1-SNAPSHOT</version>
  <name>student-records</name>
  <description>Student Records Application</description>
  <properties>
    <java.version>17</java.version>
  </properties>

  <dependencies>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-web</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-data-mongodb</artifactId>
    </dependency>

    <dependency>
      <groupId>org.springframework.boot</groupId>
      <artifactId>spring-boot-starter-validation</artifactId>
    </dependency>

    <dependency>
```

```
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-security</artifactId>
    </dependency>

    <dependency>
        <groupId>io.jsonwebtoken</groupId>
        <artifactId>jjwt-api</artifactId>
        <version>0.11.5</version>
    </dependency>

    <dependency>
        <groupId>io.jsonwebtoken</groupId>
        <artifactId>jjwt-impl</artifactId>
        <version>0.11.5</version>
        <scope>runtime</scope>
    </dependency>

    <dependency>
        <groupId>io.jsonwebtoken</groupId>
        <artifactId>jjwt-jackson</artifactId>
        <version>0.11.5</version>
        <scope>runtime</scope>
    </dependency>

    <dependency>
        <groupId>org.springframework.boot</groupId>
        <artifactId>spring-boot-starter-test</artifactId>
        <scope>test</scope>
    </dependency>

</dependencies>

<build>
    <plugins>
        <plugin>
            <groupId>org.springframework.boot</groupId>
            <artifactId>spring-boot-maven-plugin</artifactId>
        </plugin>
    </plugins>
</build>

</project>
```

Spring Boot Application Running Successfully

```
2026-09-30T18:42:32.479+05:30 INFO 12212 --- [student-records]
[main] o.s.b.w.embedded.tomcat.TomcatWebServer : Tom
cat started on port 8080 (http) with context path '/'
2026-09-30T18:42:32.486+05:30 INFO 12212 --- [student-records]
[main] c.e.s.StudentRecordsApplication : Sta
rted StudentRecordsApplication in 4.066 seconds (process running
for 4.735)
```